

“...MAESTRO MAXWELL WAS RIGHT... THESE MYSTERIOUS ELECTROMAGNETIC WAVES THAT WE CANNOT SEE WITH THE NAKED EYE. BUT THEY ARE THERE.”

Heinrich Hertz, German physicist, 1887

IN 1886, AMERICAN CHEMIST CHARLES MARTIN HALL

(1863–1914) and French scientist Paul-Louis-Toussaint Héroult (1863–1914) independently developed a technique for converting alumina—the powdery white oxide of aluminum—into **aluminum** using electrolysis.

The same year, German-born American inventor **Ottmar Mergenthaler** (1854–99) revolutionized the world of publishing when he developed the **linotype line casting machine**.

This device could set an entire line of type at a time, reducing the costs and production time of printed material. He was dubbed the “second Gutenberg” (see 1450) for his invention.

The linotype and other machines would soon run on alternating current or AC electricity (see

panel, below), thanks to American physicist **William Stanley Jr** (1858–1916). He demonstrated the **first full AC power generating system** on March 20, using it to light the town of Great Barrington, Massachusetts.

Meanwhile, German physicist Heinrich Hertz confirmed the existence of **long-wave electromagnetic radiation** (see pp.234–35)—a kind of invisible light now referred to as radio waves—that had been predicted by Scottish physicist James Clerk Maxwell in 1867.

American physicist **Henry Augustus Rowland** (1848–1901) analysed sunlight using **diffraction gratings**—glass plates or mirrors with a number of parallel lines etched onto the surface to diffract light—that he had made himself.



Lick Observatory, on Mount Hamilton, near San Jose, in California, was the first permanently occupied mountain-top observatory in the world.

THE STUDY OF LIGHT continued in 1887 as American scientists **Albert Abraham Michelson**

(1852–1931) and **Edward Morley** (1838–1923) carried out an experiment that showed that the **speed of light** is not affected by the motion of Earth through space. As predicted by James Clerk Maxwell's equations (see 1867), the measured speed of light relative to an object always remains the same, irrespective of whether the object moves head-on into a light beam, is overtaken by it, or is at any other angle from it. The **Michelson-Morley experiment** would later be taken as confirmation of German-born American physicist Albert Einstein's special theory of relativity (see 1914–15). At the time, it was seen by Michelson and Morley as a failure: they had unsuccessfully attempted to confirm the motion through the aether (the substance presumed to fill all of space, enabling light to travel through a vacuum).

Heinrich Hertz's work with radio waves led him to discover the **photoelectric effect**. He observed that a transmitter's radio waves generated sparks between two small metal spheres that were almost touching. We now know that this occurs because electromagnetic radiation knocks electrons out of a metal surface. Hertz published his

HEINRICH HERTZ (1857–94)



German physicist Heinrich Hertz is best known for the series of experiments he carried out to test James Clerk Maxwell's theories of electromagnetism. These included transmission and detection of radio waves and proving that light is a form of electromagnetic vibration. The unit of frequency—cycles per second—is called the hertz (Hz) in his honor.

observations of the phenomenon in the journal *Annalen der Physik* (*Annals of Physics*) in 1887.

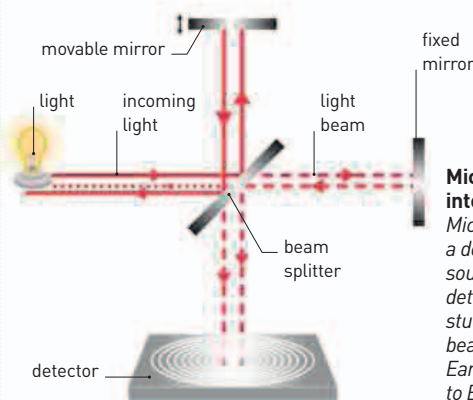
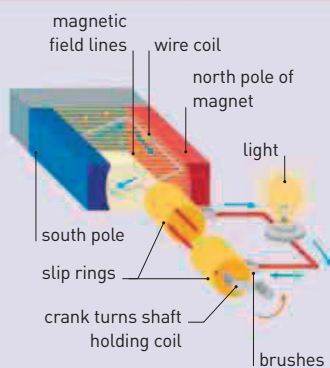
The same year, American inventor **Herman Hollerith** (1860–1929) received a patent for his **punched card tabulating machine**, which helped tabulate census statistics. This machine

was a forerunner of the electronic computer.

On the last day of the year, a refracting telescope with a 36-in (91-cm) diameter lens was completed at Lick Observatory in California. It was first used on January 3, 1888. At the time, it was the **biggest in the world**.

ALTERNATING CURRENT

When a loop of wire is rotated between the poles of a magnet, alternating electrical current is generated. The current flowing in the wire reverses repeatedly (alternates) as it turns. Main, or three-phase electricity, is generated using three coils oriented at 120 degrees to each other. Domestic power supplies most commonly alternate 50 or 60 times a second.



Michelson-Morley interferometer

Michelson and Morley built a device consisting of a light source, two mirrors, and a detector. They used it to study interference between beams of light moving with Earth and at right angles to Earth's motion.

Ottmar Mergenthaler invents the **linotype line casting machine**

March 20 William Stanley, Jr. builds the first complete system for **transmission of high-voltage alternating current**

Austro-German psychiatrist Richard von Krafft-Ebing's ***Sexual Psychopathology: A Clinical-Forensic Study*** is published

July 3 American newspaper New York Tribune becomes the first to use the **linotype**

November 11 Heinrich Hertz confirms the existence of **electromagnetic waves** predicted by Maxwell

March 3, 1887 Irish-American teacher Anne Sullivan begins teaching Helen Keller, the **first deaf and blind person to get a degree**

1887 Heinrich Hertz publishes his work on the **photoelectric effect** and **radio waves**

1887 Herman Hollerith patents his **punched card tabulating machine**

November 8, 1887 German-born American inventor Emile Berliner is granted a US patent for the **gramophone**

January 27, 1888 The **National Geographic Society** is founded in the US

1887 Albert Abraham Michelson and Edward Morley suggest that the **speed of light** is an absolute constant

1887 German physiologist Adolf Fick invents a **contact lens** that is tolerated by the human eye

January 3, 1888 The Lick **refracting telescope** becomes operational

May 1888 Nikola Tesla patents the **induction motor**